AIORI HACKATHON 3

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LifeBridge AI

AI-Powered Personalized Opportunity Intelligence Platform for Scholarships, Welfare Schemes, Internships and Fellowships

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Abstract & Problem Description

Abstract

LifeBridge AI is an AI-powered, personalized opportunity intelligence platform. It collects verified information from official sources, extracts key details from notifications, matches opportunities to each user's profile and explains every eligibility decision with supporting evidence. It also builds application checklists, tracks deadlines, offers multilingual explanations and flags suspicious or outdated listings.

Key issues

- **Scattered information:** many websites and documents
- **Confusing criteria:** eligibility, income limits, renewal
- **Missed deadlines:** incomplete applications
- **Outdated or misleading** opportunities
- **No personalized guidance,** mostly English only

Problem statement: people miss scholarships, welfare schemes, internships and fellowships because information is scattered, confusing, outdated or misleading. LifeBridge AI makes opportunities easier to discover, understand and access.

Background

Scholarships

Merit, need-based and category-specific awards with strict criteria

Welfare schemes

Central and state schemes with income limits and document rules

Internships & fellowships

Time-bound programs with changing eligibility and deadlines

In scope

- Verified, official sources only
- Eligibility matching with evidence
- Document extraction and checklists
- Multilingual explanations and alerts

Out of scope

- Submitting applications for users
- Guaranteeing selection or approval
- Trusting unverified third-party listings
- Legal or financial advice

Literature & Existing-Work Review

Source	Design lesson taken
myScheme and National Scholarship Portal [1]	Central discovery exists; add personalized, evidence-based eligibility
Lewis et al., RAG, 2020 [2]	Ground every answer in retrieved official documents
Reimers & Gurevych, SBERT, 2019 [3]	Semantic matching of user profiles to opportunity text
Xu et al., LayoutLM, 2020 [4]	Extract fields from scanned, layout-heavy notifications
Ribeiro et al., LIME, 2016 [5]	Explanations build user trust in automated decisions
Gala et al., IndicTrans2, 2023 [6]	Multilingual explanations for Indian languages

Intended contribution: verified sources + document extraction + profile matching + evidence-backed explanations + trust flags, in one platform.

Proposed Solution

1 Source collector

Gathers notifications only from verified official portals and sites

2 Document analyzer

Reads PDFs and notices; extracts criteria, limits, deadlines, documents

3 Knowledge base

Structured, versioned opportunity records with source links

4 Profile matcher

Rules plus semantic ranking match opportunities to each profile

5 Explainer & checklist

Cites evidence per decision; builds checklists in the user's language

6 Trust & alerts

Tracks deadlines; flags suspicious, expired or outdated information

Match verdicts Eligible | Likely eligible | Needs review | Not eligible

Every verdict links back to the official source text that supports it.

Optimization Proposed

Area	Basic approach	Planned optimization
Discovery	Manual search across many sites	Profile-based matching and ranking
Eligibility	Read long PDFs by hand	Extracted criteria + rule check with evidence
Freshness	Stale pages and old notices	Versioned sources, expiry and change detection
Trust	Unverified listings	Source verification and suspicious-listing flags
Access	English-only, generic guidance	Multilingual, personalized checklists

Metrics we will report

Extraction accuracy

Match precision and recall

Outdated-listing detection rate

Planned improvements, not measured gains.

Solution Architecture & Design

1. Source collection
Official portals · notifications · PDFs · refresh schedule

2. Document analysis
OCR · field extraction · criteria, dates, required documents

3. Opportunity knowledge base
Structured, versioned records · source provenance

4. Matching & eligibility engine
Profile rules · semantic ranking · verdict with confidence

5. Explanation & checklist
Evidence citations · personalized checklist · multilingual output

6. Alerts & trust monitor
Deadline reminders · outdated and suspicious flags · review queue

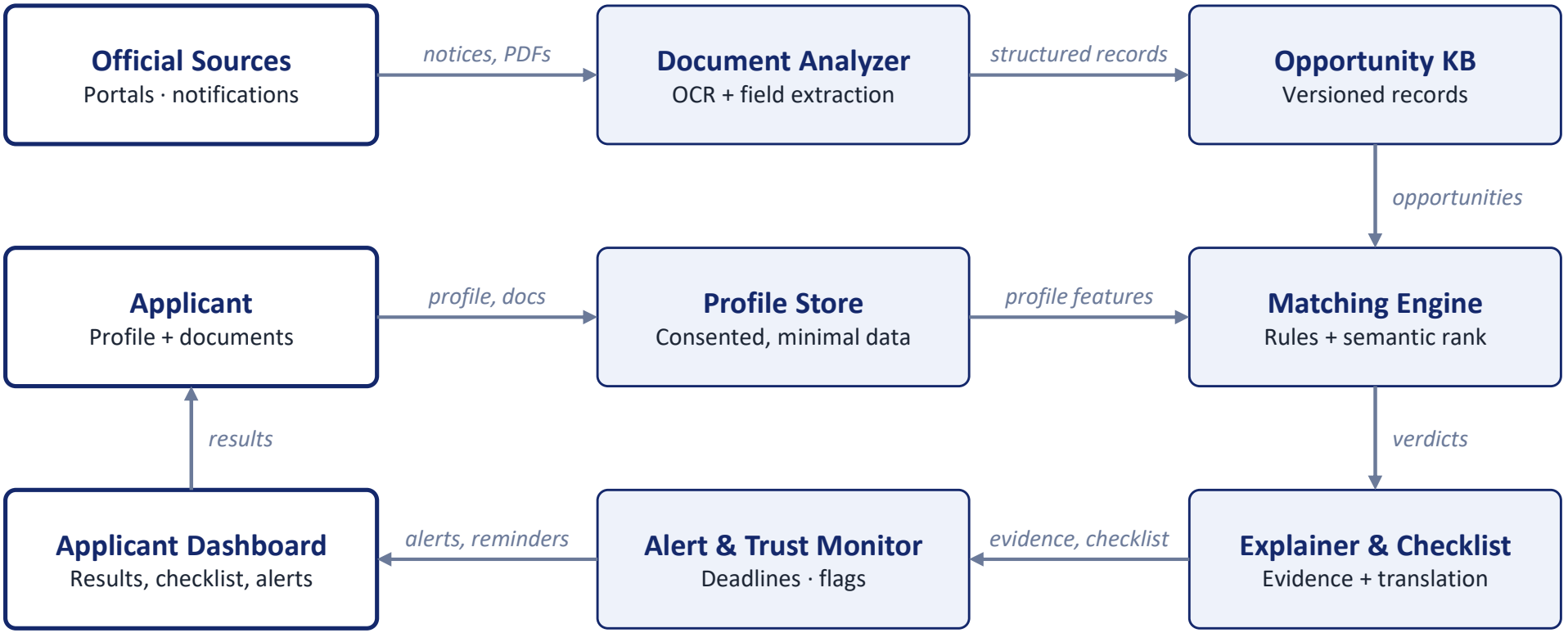
Trust controls

- Official sources only
- Evidence for each verdict
- Versioned data and rules
- Low-confidence review

Privacy controls

- Minimal profile data
- User consent first
- Encrypted storage
- Delete on request

Data Flow Diagram



Deployment Timeline

1

Research & scope

Sources, user needs, data schema

Deliverable
Requirement spec

2

Data collection

Crawl verified official portals

Deliverable
Opportunity dataset

3

Extraction

Parse notices; criteria, dates, documents

Deliverable
Field extractor

4

Matching & explain

Rules, ranking, evidence citations

Deliverable
Eligibility engine

5

App & languages

Checklists, alerts, multilingual UI

Deliverable
Working prototype

6

Test & present

Accuracy checks, user trials, demo

Deliverable
Demo + evaluation

AI & Data workstream
Extraction, matching, explanations, evaluation

Product & Platform workstream
Collectors, APIs, app, alerts, multilingual UI

Bibliography

[1] Government of India. myScheme (myscheme.gov.in) and National Scholarship Portal (scholarships.gov.in).

[2] P. Lewis et al. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.” NeurIPS 2020.

[3] N. Reimers, I. Gurevych. “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.” EMNLP-IJCNLP 2019.

[4] Y. Xu et al. “LayoutLM: Pre-training of Text and Layout for Document Image Understanding.” KDD 2020.

[5] M. T. Ribeiro, S. Singh, C. Guestrin. “‘Why Should I Trust You?’: Explaining the Predictions of Any Classifier.” KDD 2016.

[6] J. Gala et al. “IndicTrans2: Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Indian Languages.” TMLR 2023.

[7] Government of India. Digital Personal Data Protection Act, 2023.

Verify citations against the original publications before submission.

Thank You for Your Attention!

LifeBridge AI — discover, understand and access the right opportunities